

Trained



train



Large labelled
training set

Corpus-specialized
features

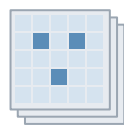
Standard



Cross-env.

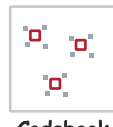


Training-free, prior work



Frozen patch
features

Standard



Codebook

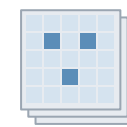
Prior-selected / aggregated features

Cross-env.



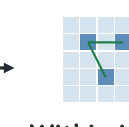
Segmentation CLS attention

Ours

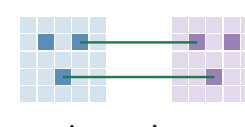


Frozen patch
features

Standard



Within-image
Balanced, matchable patch evidence



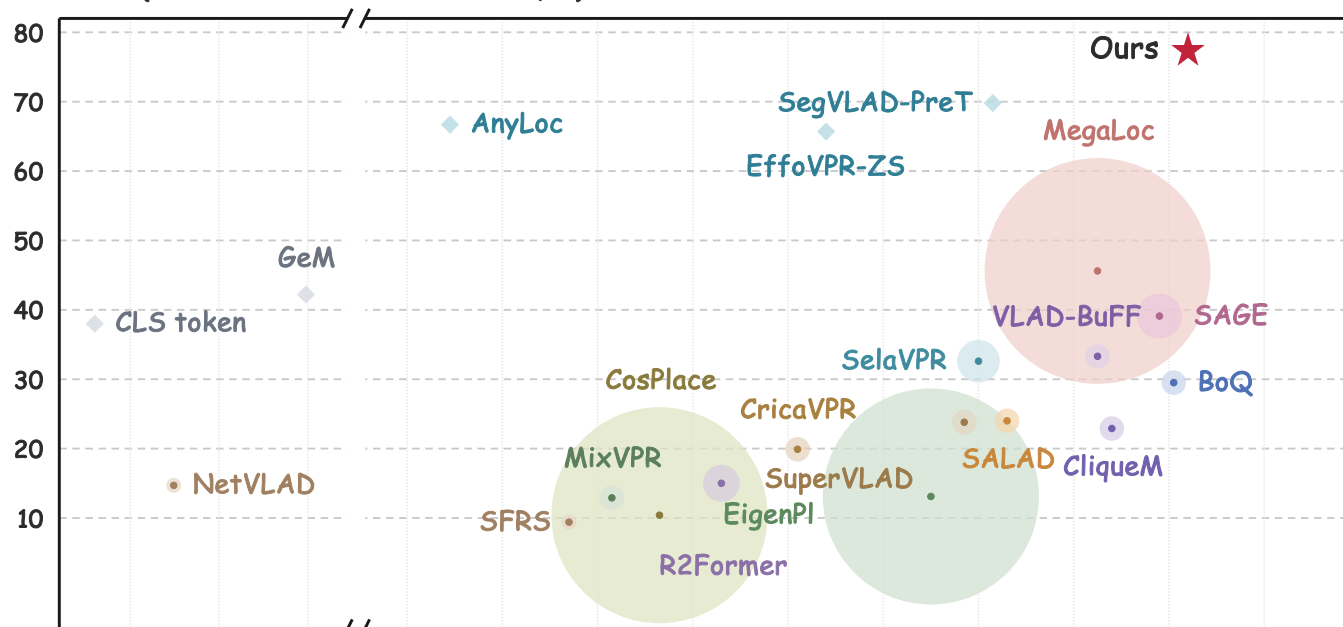
Across-image
Balanced, matchable patch evidence

Cross-env.



Training cost vs. performance

VP-Air R@1 (cross-environment benchmark, %)



Tokyo 24/7 R@1 (standard benchmark, %)

★ ours (training-free) ♦ training-free ● trained ● 1 million images ● 5 million images

Performance across 17 benchmarks

